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TERM END EXAMINATION

WINTER - 2024

PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE : CM5474-LINUX OPERATING SYSTEM

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

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| Q.1.(A) Attempt any ONE. | 02 |
| a) Enlist any 4 services provided by operating system. | |
| b) Recall the term process and its life cycle. | |
| (B) Attempt any TWO. | 08 |
| a) Describe following services provided by operating system. | |
| i) File system manipulation ii) Program Execution | |
| iii) Resource management iv) Error handling | |
| b) Demonstrate how to get current runlevel of Linux operating system and how to change the current runlevel by giving suitable example. | |
| c) Describe following components of Linux operating system. | |
| i) Kernel ii) System Library iii) System Utility | |
| Q.2.(A) Attempt any ONE. | 02 |
| a) State the use of cal and date command by giving example. | |
| b) Recall the use of following Linux command. i) echo ii) tty | |
| (B) Attempt any THREE. | 12 |
| a) Describe parent child relationship in Linux file system and enlist common directories in Linux and give their uses. | |
| b) Describe the term path and explain absolute path and relative path with suitable example. | |
| c) Demonstrate use of grep command with its any two options. | |
| d) Demonstrate use of following Linux commands : i) cd ii) mkdir iii) rmdir iv) ls | |
| Q.3.(A) Attempt any TWO. | 08 |
| a) Develop a Shell script that prompts user to enter name of the directory to be created if directory already exist prompt the message "Directory already exist". If directory does not exist. Create the new directory and prompt the message "Directory created successfully". | |
| b) Explain the case statement in Shell script with suitable example. | |
| c) Explain command line arguments in Shell script with example. | |



- (B) Attempt any **TWO**. 08
- a) Describe logical operators and expr commands with its examples.
 - b) Design a Shell script to print Fibonacci series.
 - c) Design a Shell script to find weather entered number is divisible by 5 & 9.
- Q.4.(A)** Attempt any **ONE**. 02
- a) State any four duties of system administrator.
 - b) Write procedure for system administrator login.
- (B) Attempt any **TWO**. 08
- a) Describe task scheduling using cron.
 - b) Demonstrate the use of following Linux commands :
 - i) useradd ii) userdel iii) groupadd iv) groupdel
 - c) Construct Linux command for :
 - i) shutdown the Linux system at 5 am.
 - ii) shutdown the Linux system after 20 minutes
 - iii) cancel the scheduled shutdown
 - iv) shutdown the system immediately
- Q.5.(A)** Attempt any **ONE**. 02
- a) State the function of find command.
 - b) Recall the term RAID.
- (B) Attempt any **TWO**. 08
- a) Demonstrate the use of tar and cpio command to create backup.
 - b) Define Backup and explain the need of Backup.
 - c) Demonstrate the use of df and du command with its any two options.
- Q.6.(A)** Attempt any **ONE**. 02
- a) Recall the term DHCP server.
 - b) Recall the term intrusion detection.
- (B) Attempt any **TWO**. 08
- a) Explain the working of firewall and demonstrate iptables based firewall.
 - b) Demonstrate use of following Linux commands : i) netstat ii) route iii) ping iv) ifconfig
 - c) Describe NIS and NFS in detail.
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